

Laravel Category System (Many-to-Many Hierarchical) — Full Structure

Below is a fully prepared structure for: - Migrations - Models - Seeders - Controllers - Admin Panel UI (Blade) - Product Binding - Showing Categories on Product Page

All relationships allow **multiple parents per category**.

1. Migrations

categories table

```
Schema::create('categories', function (Blueprint $table) {
    $table->id();
    $table->string('name');
    $table->string('slug')->unique();
    $table->timestamps();
});
```

category_parent (pivot)

```
Schema::create('category_parent', function (Blueprint $table) {
    $table->id();
    $table->foreignId('category_id')->constrained('categories')->
    >cascadeOnDelete();
    $table->foreignId('parent_id')->constrained('categories')->
    >cascadeOnDelete();
});
```

product_categories pivot (already exists – included for reference)

```
Schema::create('product_categories', function (Blueprint $table) {
    $table->id();
    $table->foreignId('product_id')->constrained()->cascadeOnDelete();
    $table->foreignId('category_id')->constrained()->cascadeOnDelete();
    $table->timestamps();
});
```

2. Models

Category.php

```
class Category extends Model
{
    protected $guarded = ['id'];

    public function parents()
    {
        return $this->belongsToMany(Category::class, 'category_parent',
'category_id', 'parent_id');
    }

    public function children()
    {
        return $this->belongsToMany(Category::class, 'category_parent',
'parent_id', 'category_id');
    }

    public function childrenRecursive()
    {
        return $this->children()->with('childrenRecursive');
    }

    public function products()
    {
        return $this->belongsToMany(Product::class, 'product_categories');
    }
}
```

3. Seeder for Category Tree

```
class CategorySeeder extends Seeder
{
    public function run()
    {
        $men = Category::create(['name' => 'Men', 'slug' => 'men']);
        $women = Category::create(['name' => 'Women', 'slug' => 'women']);

        $summer = Category::create(['name' => 'Summer', 'slug' => 'summer']);
        $clothing = Category::create(['name' => 'Clothing', 'slug' =>
'clothing']);

        $summer->parents()->attach([$men->id, $women->id]);
        $clothing->parents()->attach($summer->id);
    }
}
```

```
}  
}
```

4. Admin CategoryController

```
class CategoryController extends Controller  
{  
    public function index()  
    {  
        $categories = Category::with('parents')->paginate(20);  
        return view('admin.categories.index', compact('categories'));  
    }  
  
    public function create()  
    {  
        $categories = Category::all();  
        return view('admin.categories.create', compact('categories'));  
    }  
  
    public function store(Request $request)  
    {  
        $data = $request->validate([  
            'name' => 'required',  
            'slug' => 'required|unique:categories',  
            'parent_ids' => 'array',  
        ]);  
  
        $category = Category::create($data);  
  
        if ($request->parent_ids)  
            $category->parents()->sync($request->parent_ids);  
  
        return redirect()->route('admin.categories.index');  
    }  
  
    public function edit(Category $category)  
    {  
        $categories = Category::all();  
        return view('admin.categories.edit', compact('category',  
'categories'));  
    }  
  
    public function update(Request $request, Category $category)  
    {  
        $data = $request->validate([  
            'name' => 'required',  
            'slug' => 'required|unique:categories,slug,' . $category->id,  
            'parent_ids' => 'array',  
        ]);  
    }  
}
```

```

    });

    $category->update($data);
    $category->parents()->sync($request->parent_ids);

    return redirect()->route('admin.categories.index');
}

public function destroy(Category $category)
{
    $category->delete();
    return back();
}
}

```

5. Admin Panel - Add Category UI

create.blade.php

```

<form method="POST" action="{{ route('admin.categories.store') }}">
    @csrf

    <label>Name</label>
    <input type="text" name="name" required>

    <label>Slug</label>
    <input type="text" name="slug" required>

    <label>Select Parents (Multiple Allowed)</label>
    <select name="parent_ids[]" multiple>
        @foreach($categories as $cat)
            <option value="{{ $cat->id }}">{{ $cat->name }}</option>
        @endforeach
    </select>

    <button type="submit">Create</button>
</form>

```

6. Binding Products with Categories

In ProductController:

```

$product->categories()->sync($request->category_ids);

```

Product Form UI:

```
<label>Select Categories</label>
<select name="category_ids[]" multiple>
    @foreach($categories as $cat)
        <option value="{{ $cat->id }}">{{ $cat->name }}</option>
    @endforeach
</select>
```

7. Frontend: Showing Categories in Product Page

Example (product detail page):

```
<h4>Categories:</h4>
<ul>
    @foreach($product->categories as $category)
        <li>{{ $category->name }}</li>
    @endforeach
</ul>
```

8. Frontend: Show Full Recursive Tree (Sidebar)

```
@foreach($categories as $cat)
    <div>
        <strong>{{ $cat->name }}</strong>
        @if($cat->childrenRecursive->count())
            <ul>
                @include('components.category_tree', ['children' => $cat->childrenRecursive])
            </ul>
        @endif
    </div>
@endforeach
```

components/category_tree.blade.php

```
@foreach($children as $child)
    <li>
        {{ $child->name }}
        @if($child->childrenRecursive->count())
            <ul>
                @include('components.category_tree', ['children' => $child->childrenRecursive])
            </ul>
        @endif
    </li>
</foreach>
```

```
        </ul>
      @endif
    </li>
  @endforeach
```

✓ Completed:

- Migrations
- Models
- Seeders
- Controllers
- Admin Panel UI
- Product binding
- Frontend category display system

Agar chaho to next step me **API version, Vue/React admin panel**, ya **SEO friendly breadcrumb system** bhi bana dun.